

Aadarsh S U

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SUMMARY

Software Engineer with ~2 years of experience building backend systems, AI-powered automation pipelines, and developer tooling. Specialized in Python-based architectures using Django and FastAPI, with hands-on expertise in multi-agent AI systems, RAG pipelines, and cloud-integrated workflows. Strong track record of delivering measurable efficiency gains in production environments.

EXPERIENCE

HashedIn by Deloitte

On-Site - Chennai, India

Software Engineer 1

Nov 2025 – Present

- Built an **Agentic AI system (Django)** that auto-generates XML, XSLT, SQL, and Oracle Fast Formulae, cutting manual generation time by **~70%** and reducing formula error rates by **~40%** through **Azure OpenAI** and **Anthropic model** integration.
- Designed multi-agent orchestration translating natural language into 10+ Oracle formula types with end-to-end accuracy, backed by **Azure Cosmos DB** retrieval and rule-based validation.
- Optimized bulk upload performance by re-engineering ingestion logic to batch queries against **Azure Cosmos DB**, significantly reducing Request Unit (RU) consumption and improving throughput.
- Implemented **RAG-based retrieval** for company-specific QnA & contextual validation, cutting domain query resolution time by **~60%**.
- Engineered end-to-end Oracle eText and RTF template generation – from POC through full-scale production rollout—owning automated conditional formatting and logic validation to ensure production-ready output.
- Solely owned the **Workday Offering integration** – drove all feature changes across XSLT generation, multi-provider **LLM routing**, and multi-format file handling.
- Built and shipped new feature APIs as requirements evolved, backed by **pytest** test suites achieving **>80% code coverage**; containerized services with **Docker** for consistent deployment across environments.

Microchip Technology

On-Site - Chennai, India

Engineer 1 - Corporate Applications

Jan 2024 – Mar 2025

- Built a **Python + OpenCV** tool to extract pin names from schematics, adopted by global engineering teams for design validation.
- Built a custom **network testing framework in Python** for proprietary hardware, improving test efficiency and reducing manual effort by **30%**.
- Executed **functional and protocol tests** (IEEE PTP, ACL) for Linux-based wired network products using custom Python scripts.
- Developed a **6-screen GUI application** with modular architecture, packaged as a standalone executable for cross-team deployment.
- Designed an **embedded C solution** enabling Ethernet communication on entry-level microcontrollers, achieving a **25%** improvement in data transmission speed.

PROJECTS

- Multi-Agent Code Analysis:** Built a **LangGraph-based multi-agent system** that auto-generates role-specific documentation. Features **FastAPI** real-time pipeline with mid-run steering (~50% fewer re-runs), **pgvector** semantic code retrieval (~80% accuracy), and a **Streamlit** monitoring dashboard with codebase Q&A.
- WhisperQL:** Full-stack app (**FastAPI** + **Streamlit**) that converts **plain-English** questions to executable **SQL** via multi-LLM integration (**Gemini**, **Claude**, **GPT**) with auto schema introspection and a pre-execution safety gate.
- Email Automation Script:** Developed a **Python** script using **Selenium WebDriver** to automate email sending for a college club event, successfully delivering over **500** invitation emails to participants.
- Hostel Management System:** Built a web-based student record management system using **HTML**, **CSS**, and **JavaScript** with complete **CRUD** operations, integrating a relational database for efficient data storage, retrieval, and management.
- IoT Smog Prediction System:** End-to-end environmental monitoring with NodeMCU, custom PCB, and **ThingSpeak** cloud integration. Applied predictive modeling (**LPC**) for real-time trend analysis and alerting.

SKILLS

- Languages:** Python, Java, JavaScript, PHP, SQL, C
- Frameworks:** FastAPI, Django, Streamlit
- Database:** PostgreSQL, Azure CosmosDB
- Generative AI:** Langchain, Langgraph, Autogen, RAG, OpenAI, Anthropic
- Version Control:** Git
- Web Technologies:** HTML, CSS
- Testing:** Pytest

EDUCATION

Thiagarajar College of Engineering

Madurai, India

Bachelors of Engineering in Electronics and Communication Engineering

CGPA 9.61 | Sep 2020 – May 2024

CERTIFICATIONS

- Azure AI Fundamentals - Microsoft
- Django & FastAPI – The complete course (Beginner + Advanced + AI) - Udemy
- Building AI Agents & Agentic AI system via Microsoft Autogen – Udemy
- Taming Big Data with Apache Spark 4 and Python - Hands On!
- Claude with Amazon Bedrock - Anthropic